

Challenge (Habanero)

- Q1.** In a game, a player's score can be represented as $x \cdot y$, where x and y are integers. If $x = -3$ and $y = 4$, what is the score?
(A) -12
(B) 12
(C) -3
(D) 4
(E) 0
- Q2.** A coding error occurs every 5^{th} line of code. If a programmer writes 23 lines of code, how many errors will occur?
(A) 4
(B) 5
(C) 3
(D) 2
(E) 1
- Q3.** The number of players in an online game follows the pattern: 2, 5, 8, 11, 14, What is the next number in the pattern?
(A) 15
(B) 17
(C) 18
(D) 20
(E) 21
- Q4.** In a scoreboard, a team's score is represented as $-2x + 5$, where x is an integer. If the team's score is -3 , what is the value of x ?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q5.** A computer program uses 3 integers to generate a code: x, y, z . If $x = -2, y = 4, z = -1$, what is the value of $x + y + z$?
(A) 1
(B) 3
(C) 0
(D) -1
(E) -3
- Q6.** The number of bytes used by a game is given by the formula $2^n - 1$, where n is an integer. If $n = 5$, how many bytes are used?
(A) 31
(B) 32
(C) 33
(D) 63
(E) 65
- Q7.** A programmer uses the pattern 1, 4, 7, 10, 13, ... to generate a sequence of numbers. What is the 6^{th} term in the sequence?
(A) 15
(B) 16
(C) 17
(D) 19
(E) 20
- Q8.** The score of a player in a game is represented as $x - 2y$, where x and y are integers. If $x = 5$ and $y = -3$, what is the score?
(A) 11
(B) 9
(C) 7
(D) 5
(E) 3

Open Ended Questions (FRQ)

- Q9.** A game uses the formula $3x - 2y$ to calculate the score, where x and y are integers. If the score is -5 , and $x = -2$, find the value of y . Show your work and explain your reasoning.
- Q10.** A computer program generates a sequence of numbers using the formula $2n + 1$, where n is an integer. If the sequence starts with 3, 5, 7, ..., find the 10^{th} term in the sequence. Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Encourage students to use visual aids to help with problem-solving.
- Tip 2: Remind students to check their work for accuracy.
- Tip 3: Provide feedback that guides students towards the correct solution.